

PRIMES NEAR CARMICHAEL NUMBERS: A TRADE-OFF THEOREM AND ITS BARRIERS

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ABSTRACT. Let $C(X)$ denote the number of Carmichael numbers up to X . We prove an unconditional trade-off: if $C(X) - C(X/2) \geq X^\theta$, then some Carmichael number $m \in (X/2, X]$ has a prime within $m^{1.23-\theta+\varepsilon}$ of it. Consequently any counting bound $C(X) \geq X^{0.706}$ would yield primes nearer to (some) Carmichael numbers than the Baker–Harman–Pintz exponent 0.525 guarantees near arbitrary points, with no new prime-gap input; under the Erdős–Pomerance abundance conjecture the exponent drops to $0.23 + \varepsilon$, and under the Lindelöf hypothesis to ε . We complement this with two negative results. First, a barrier: since $C(X) \leq X^{1-(1+o(1)) \log \log \log X / \log \log X}$ unconditionally, no pigeonhole of this type — under any abundance assumption and any variance-strength bound on primeless neighborhoods, including those available under the Riemann Hypothesis — can place a prime within $\exp((1 - \varepsilon) \log \log \log m \cdot \log m / \log \log m)$ of a Carmichael number; in particular polylogarithmic proximity is out of reach for the method. Second, an algorithmic accounting: a deterministic prime-finding algorithm whose correctness certificate is a counting inequality of this type must test $X^{1+o(1)}$ candidates, so it cannot compete with the trivial $X^{0.525+o(1)}$ scan, let alone the $X^{1/2+o(1)}$ record of Lagarias and Odlyzko. On the positive side, the Alford–Granville–Pomerance equidistribution theorem underlying the companion construction is itself a prime-existence theorem for a thin explicit set: it yields, unconditionally and effectively, a deterministic algorithm finding a prime in $(N/2, N]$ in time $N^{7/12+o(1)}$ — to our knowledge new, and the third-fastest proven method. Its exponent is $1 - 1/A$ for any admissible zero-density exponent A , hence $N^{1/2+\varepsilon}$ under the density hypothesis and never below $N^{1/2}$, since $A \leq 2$ is impossible. Empirically, Carmichael numbers are not measurably closer to primes than random odd numbers of the same size, although the primes near them are confined to a sparse, predictable set of admissible offsets.

1. INTRODUCTION

A companion paper [3] constructs, deterministically and unconditionally, a Carmichael number exceeding a given bound n in time quasi-polylogarithmic in n . By contrast, the fastest deterministic algorithm known to produce a *prime* exceeding n runs in time $n^{1/2+o(1)}$ [8, 10]. This prompts a natural question: *must primes lie close to Carmichael numbers?* If Carmichael numbers — which we can find quickly — were provably accompanied by nearby primes, one could find primes by searching their neighborhoods, and any proximity exponent below $1/2$ would beat the record.

This note is an after-action report on the most natural proof strategy, a pigeonhole against prime-gap moments: what it delivers, and where it is stopped. Write $p_1 < p_2 < \dots$ for the primes, $d_n = p_{n+1} - p_n$, and let

$$E_H(X) = \{x \in (X/2, X] : \text{no prime in } [x - H, x + H]\}.$$

Write $C(X)$ for the number of Carmichael numbers up to X , and $\ell_2 = \log \log X$, $\ell_3 = \log \log \log X$.

Our inputs are classical and recent bounds on both sides of the pigeonhole. For the gaps: Selberg [12] proved, under the Riemann Hypothesis, that $\sum_{p_n \leq x} d_n^2 \ll x \log^3 x$; unconditionally, Heath-Brown [7] obtained the exponent $23/18$, improved through work of Peck and Maynard to $5/4$, and by Stadlmann

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[13] to

$$(1) \quad \sum_{p_n \leq x} d_n^2 \ll_{\varepsilon} x^{1.23+\varepsilon},$$

under the Lindelöf hypothesis Yu obtained exponent $1 + \varepsilon$ (see the introduction of [13] for this history). For the Carmichael counts: $C(X) \geq X^{1/3}$ for large X (Harman [6], refining Alford–Granville–Pomerance [1]), while unconditionally

$$(2) \quad C(X) \leq X^{1-(1+o(1))\ell_3/\ell_2}$$

(Pomerance [11]), and the standard conjecture of Erdős and Pomerance (see [4, 5]) asserts that (2) is an equality.

2. THE TRADE-OFF THEOREM

Lemma 2.1. *For $1 \leq H \leq X/8$ and every $\varepsilon > 0$,*

$$|E_H(X)| \leq \frac{1}{2H} \sum_{p_n \leq 2X} d_n^2 \ll_{\varepsilon} \frac{X^{1.23+\varepsilon}}{H}.$$

Under the Riemann Hypothesis, if additionally $H \geq 12 \log X$, then $|E_H(X)| \ll X \log^3 X/H$.

Proof. If $x \in E_H(X)$ then the consecutive primes $p_n < x < p_{n+1}$ that bracket x satisfy $p_{n+1} - x > H$ and $x - p_n > H$, so $d_n > 2H$; moreover $p_n < X$ and $p_{n+1} \leq x + d_n \leq 2X$ once $d_n \leq X$ (gaps below $2X$ of length $> X$ do not exist for large X by Bertrand’s postulate). Each such gap contains fewer than d_n integers x , whence $|E_H(X)| \leq \sum_{d_n > 2H, p_n \leq 2X} d_n \leq (2H)^{-1} \sum_{p_n \leq 2X} d_n^2$, and (1) gives the first bound.

For the second: Selberg’s theorem [12] gives, under RH, $\int_{X/2}^X (\psi(x+H) - \psi(x) - H)^2 dx \ll HX \log^2 X$ (we use the form with a $\log^3$ -safe constant; any bound $\ll HX \log^{O(1)} X$ serves). If $x \in E_H(X)$ then the interval $(x, x+H]$ contains no primes, only prime powers p^j , $j \geq 2$, of which there are at most $HX^{-1/2} + 2$, each contributing at most $\log 2X$ to ψ ; thus $\psi(x+H) - \psi(x) \leq (HX^{-1/2} + 2) \log 2X \leq H/2$ for $H \geq 12 \log X$ and X large, so the integrand is at least $H^2/4$ on $E_H(X)$. Comparing measures gives $|E_H(X)| \ll X \log^3 X/H$. $\square$

Theorem 2.2. *Let $0 < \theta \leq 1$ and $\varepsilon > 0$, and suppose $C(X) - C(X/2) \geq X^\theta$ for all large X . Then there are infinitely many Carmichael numbers m admitting a prime p with*

$$|p - m| \leq m^{1.23-\theta+\varepsilon}.$$

Proof. Set $H = X^{1.23-\theta+\varepsilon}$ (if the exponent is ≥ 1 the claim is trivial by Bertrand’s postulate, so assume it is < 1). By Lemma 2.1, $|E_H(X)| \ll X^{1.23+\varepsilon/2}/H = X^{\theta-\varepsilon/2} < X^\theta \leq C(X) - C(X/2)$ for large X . Hence some Carmichael $m \in (X/2, X]$ lies outside $E_H(X)$ and has a prime within $H = X^{1.23-\theta+\varepsilon} \leq (2m)^{1.23-\theta+\varepsilon} \leq m^{1.23-\theta+2\varepsilon}$ for large X ; taking $X = 2^j \rightarrow \infty$ produces infinitely many such m , and relabelling ε finishes. $\square$

Corollary 2.3 (unconditional threshold). *If $C(X) - C(X/2) \geq X^\theta$ with $\theta > 1.23 - 0.525 = 0.705$, then infinitely many Carmichael numbers have primes within $m^{1.23-\theta+\varepsilon}$, closer than the exponent 0.525 that Baker–Harman–Pintz [2] guarantees for arbitrary points.*

The current record is $\theta = 1/3$ [6], so Corollary 2.3 is presently empty: for $\theta \leq 0.705$ the theorem returns exponents worse than 0.525. Its content is a *reduction*: closing the gap from 0.33 to 0.706 in Carmichael counting — a problem entirely on the composite side, where the Alford–Granville–Pomerance machinery keeps improving — would automatically produce primes abnormally close to Carmichael numbers, with prime-gap technology frozen at (1).

Corollary 2.4 (under abundance). *Assume $C(X) - C(X/2) \geq X^{1-\delta}$ for every $\delta > 0$ (a weak form of the Erdős–Pomerance conjecture). Then for every $\varepsilon > 0$ there are infinitely many Carmichael m with a prime within $m^{0.23+\varepsilon}$. Under the Lindelöf hypothesis the exponent improves to ε , for every $\varepsilon > 0$.*

Corollary 2.5 (under RH and abundance). *Assume RH and the asymptotic form $C(X) - C(X/2) \geq X \exp(-(1+\varepsilon)\ell_3 \log X/\ell_2)$ of the Erdős–Pomerance conjecture. Then infinitely many Carmichael m have a prime within $\exp((1+2\varepsilon) \log \log \log m \cdot \log m / \log \log m) = m^{o(1)}$.*

Proof. Take $H = \log^3 X \cdot \exp((1+\varepsilon)\ell_3 \log X/\ell_2)$ in the RH part of Lemma 2.1. $\square$

3. TWO BARRIERS

The exponents of Corollaries 2.4–2.5 are the floor of the method.

Proposition 3.1 (pigeonhole barrier). *Let $B(X, H)$ be any provable upper bound for $|E_H(X)|$ of the shape $B(X, H) \geq X(\log X)^{-A}$ for fixed A whenever $H \leq (\log X)^A$ — as is the case for every bound derived from a variance (second-moment) estimate for primes in short intervals, including Selberg’s RH bound. Then the inequality $B(X, H) < C(X)$ required by the pigeonhole fails for all such H , unconditionally: by (2), $C(X) \leq X^{1-(1+o(1))\ell_3/\ell_2} < X(\log X)^{-A}$ for every fixed A and large X . Consequently no pigeonhole against Carmichael counts — with any strength of abundance input, since (2) is an upper bound — can certify a prime within polylogarithmic distance of a Carmichael number; the least H it can certify is $\exp((1-o(1))\ell_3 \log X/\ell_2)$, matching Corollary 2.5.*

Proof. Immediate from (2) and the displayed shapes: for $H \leq (\log X)^A$ one has $B(X, H) \geq X(\log X)^{-A} > C(X)$, so the pigeonhole cannot close. The floor statement follows by solving $X \log^{O(1)} X/H < X^{1-(1+o(1))\ell_3/\ell_2}$ for H . $\square$

Thus the interesting regime of Theorem 2.2 is polynomial proximity m^c ; polylogarithmic proximity is denied territory for counting methods, and entry requires either gap-tail bounds far beyond second moments (Cramér-model strength: the obstruction identified by the Polymath project [10]), or structural information about Carmichael positions, such as equidistribution of subset products of the [1] prime pool in short intervals, which lies beyond current exponential-sum technology.

The second barrier concerns the intended algorithmic use.

Proposition 3.2 (cost of counting certificates). *Consider a deterministic algorithm which outputs a prime by testing candidates drawn from R neighborhoods of radius H centred at points of $(X/2, X]$ (Carmichael or otherwise), and whose correctness proof is the counting inequality $R > B(X, H)$ with B a provable bound for $|E_H(X)|$ as above. In the worst case the algorithm performs at least $B(X, H) \cdot H/\log X$ primality tests. With the unconditional bound (1) this is $\gg X^{1.23-o(1)}$; with Selberg’s RH bound it is $\gg X(\log X)^2$. In both cases the cost exceeds X , so such an algorithm is slower than the trivial scan of a single interval of length $X^{0.525}$ [2], and far slower than the $X^{1/2+o(1)}$ algorithm of Lagarias–Odlyzko [8].*

Proof. The certificate guarantees only that the union of the R neighborhoods meets the primes; it identifies no neighborhood in particular, and any $R-1$ of them may be barren, since B is the best available bound on how many barren centres exist and R must exceed it. An adversarial configuration consistent with the certificate places the guaranteed prime among the last candidates examined, whichever order the algorithm chooses, so the worst case examines a constant fraction of the $\asymp RH/\log X$ prime-candidates in the union: at least $B(X, H) H/\log X$ tests. The two instantiations are $B \cdot H \gg X^{1.23-o(1)}$ and $B \cdot H \gg X \log^3 X$. $\square$

Proposition 3.2 is a strong negative for the motivating program: *no* certified-by-counting neighborhood search — around Carmichael numbers or any other quickly-constructible markers — can approach the $X^{1/2}$ record, because the certificate itself forces super-linear work. A competitive algorithm must certify success pointwise (an interval theorem such as [2]), or by computation (the analytic method [8]), or by the kind of parity/algebraic certificate proposed by the Polymath project [10], whose Conjecture 1.1 remains the door: deciding in time $N^{1/2-c}$ whether an interval of length $N^{1/2+c}$ contains a prime would break the barrier. We record one quantitative remark on the obstruction identified there. The Polymath method computes, in time $N^{1/2-c+o(1)}$, the parity of the number of primes in a window in every residue class modulo every prime $r \leq N^{c/10}$; it wins immediately unless every such class count is even. The following shows that trivial counting cannot exclude that conspiracy.

Lemma 3.3 (even-pairing is cheap to sustain). *Let $p_1 < \dots < p_s$ be primes in an interval of length $\leq N$, and suppose that for every prime $r \leq R$ every residue class modulo r contains an even number of the p_i . Then $\pi(R) \leq s \log_2 N$.*

Proof. Fix r . Pairing the p_i within classes gives at least $s/2$ disjoint pairs with $r \mid p_i - p_j$. Summing over $r \leq R$ counts at least $(s/2)\pi(R)$ incidences $(\{p_i, p_j\}, r)$ with $r \mid p_i - p_j$; each of the at most $s^2/2$ pairs contributes at most $\omega(p_i - p_j) \leq \log_2 N$ incidences. Hence $(s/2)\pi(R) \leq (s^2/2) \log_2 N$. $\square$

A window of length $N^{1/2+c}$ has $s \leq N^{1/2+c}$ primes, so Lemma 3.3 only forbids the conspiracy for $R \gg s \log N$ — vastly beyond the $N^{c/10}$ budget the method can afford. Ruling it out below that range is a question about primes in arithmetic progressions *within short intervals*, where the conductor of the associated L -functions consumes exactly the gain: the same wall as Proposition 3.1, seen from a third side.

4. THE AGP PIGEONHOLE AS A PRIME-FINDING THEOREM

The negative results above concern primes *near* Carmichael numbers. We close the circle with a positive observation about the machinery itself: the theorem of Alford–Granville–Pomerance that supplies the prime pool for the companion construction ([1], Theorem 3.1; quoted as Lemma 2.2 of [3]) is, read directly, a prime-existence statement for a thin explicit set, and hence a deterministic prime-finding algorithm. We have not found this observation in the literature.

Fix $B < 5/12$ and let $D_B, z_3(B)$ be the effective constants of [1], Theorem 3.1. Given x , set $T = \lceil D_B + 2 + \log_2 \log x \rceil$ and let L be the product of the first T primes exceeding $64T/(1-B)$. Then L is squarefree, its prime factors lie below $x^{(1-B)/2}$, $\sum_{q|L} 1/q \leq (1-B)/32$, and $L \leq x^B$ for large x , so the theorem applies: some $k \leq x^{1-B}$ with $(k, L) = 1$ satisfies

$$\#\{d \mid L : dk + 1 \leq x \text{ prime}\} \geq \frac{2^{-D_B-2}}{\log x} 2^T \geq 1.$$

Thus the explicit set $S_x = \{dk + 1 \leq x : d \mid L, k \leq x^{1-B}, (k, L) = 1\}$, of cardinality at most $2^T x^{1-B} = x^{1-B+o(1)}$, provably contains a prime, and enumerating it with the AKS test finds one in deterministic time $x^{1-B+o(1)}$.

As stated, the pigeonhole does not control the *size* of the primes it produces, so it finds a prime cheaply rather than a prime exceeding a given bound. The dyadic form needed for prime-finding proper follows, however, directly from the actual equidistribution theorem of [1] (their Theorem 3.1), which is stronger than the pigeonhole consequence quoted above: it is a *two-sided* relative-error asymptotic,

$$\left| \pi(y; d, a) - \frac{\pi(y)}{\varphi(d)} \right| \leq \varepsilon \frac{\pi(y)}{\varphi(d)},$$

pointwise for every $d \leq x^{1/A-\delta}$ not divisible by a member of a set of at most $D_{\varepsilon,\delta}$ bad integers, valid for all $y \geq dx^{1-1/A+\delta}$, where A is any admissible zero-density exponent ([1], eq. (3.1)); [1] note that their proof of this statement is effective. Subtracting the asymptotic at $y = dx^{1-B}$ and at $y = x/2$ (both in range when $B \leq 1/A - \delta$ and $d \geq 0.55x^B$) counts primes $p \equiv 1 \pmod d$ with $p \in (x/2, dx^{1-B}]$, hence with cofactor $k = (p-1)/d \leq x^{1-B}$; summing over the divisors of L in the dyadic block $(0.55x^B, x^B]$ that avoid the bad set, and pigeonholing over $k \leq x^{1-B}$, yields a shift k for which some $dk + 1 \in (x/2, x]$ is prime, provided the block contains $\gg 2^D \log x$ good divisors. That supply is arranged constructively: take $T = O(D + \log \log x)$ primes in an interval $(Q, 2Q)$ with $Q^{T/2} \approx x^{1/A-\delta}$; the $\binom{T}{T/2}$ subset products of size $T/2$ spread over $T/2$ dyadic blocks, so some block receives $\geq 2^T/T^2$ of them, and discarding divisors met by the bad set costs a further factor 2^{-D} at most. (The hypotheses $(k, L) = 1$ and $\sum_{q|L} 1/q$ small in the quoted pigeonhole are needed for the Carmichael application, not here.) This proves:

Theorem 4.1 (unconditional). *For every fixed $B < 5/12$ there is a deterministic algorithm that, given N , outputs a prime in $(N/2, N]$ in time $N^{1-B+o(1)}$; in particular in time $N^{7/12+o(1)}$. The constants are effectively computable, and the exponent is $1 - 1/A + o(1)$ for any admissible zero-density exponent A in (3.1) of [1].*

Three remarks. First, the exponent slots into the known ladder as follows: Lagarias–Odlyzko $N^{1/2+o(1)}$ [8]; the Baker–Harman–Pintz interval scan $N^{0.525+o(1)}$ [2]; Proposition 4.1 at $N^{7/12} = N^{0.5834}$; the Meissel–Lehmer binary search and the Heath–Brown $a^3 + 2b^3$ scan, both $N^{2/3+o(1)}$ (see [10]). The new entry is the only one of a multiplicative species — a union of $\log^{O(1)} N$ values $d \mid L$ sampled along N^{1-B} shifts — and the only one whose exponent is a dial rather than a constant. Second, the dial: the mechanism $B < 1/A$ is stated by [1] themselves (“if $A \in \mathcal{A}$, then all $B < 1/A$ are in $\mathcal{B}$ ”), with Huxley’s $A = 12/5$ giving $5/12$. The Guth–Maynard improvement of the zero-density exponent to $A = 30/13$ would lift B to $13/30$ and the exponent of Theorem 4.1 to $N^{17/30+o(1)}$ — their short-interval exponent, reached by a purely multiplicative route — provided $30/13$ is admissible in the hybrid q -aspect sense of (3.1) of [1], which is proved there for the t -aspect and remains to be checked for conductors. Progress toward the density hypothesis ($A \rightarrow 2^+$) pushes the exponent to $N^{1/2+\varepsilon}$, at which point an elementary scan matches the analytic record — and there it stops, unconditionally: [1] observe that (3.1) cannot hold for any $A \leq 2$, because $L(s, \chi)$ really does have $\asymp Td \log(Td)$ zeros up to height T on the critical line. The infimum of the exponent $1 - 1/A$ over *all possible* zero-density inputs is therefore exactly $1/2$: the route can match the record in the limit and can never beat it. Nor can level-of-distribution technology rescue it. The divisor family $\{d \mid L\}$ is necessarily sparse ($\tau(L) = N^{o(1)}$), so the pigeonhole needs *pointwise* control with bounded exceptions — zero-density territory, capped as above. The theorems that go beyond level $1/2$ bound errors absolutely, at the scale of a dense family: this includes the well-factorable theorem of Bombieri–Friedlander–Iwaniec, which does apply here — indicator weights on the divisors of a smooth L are the archetype of well-factorable weights — but its error term $x/\log^A x$ swamps our thin main term $x^{1-B+o(1)}$; and a relative-error average theorem for thin families at level beyond $1/2$ would, applied to a singleton family, exceed what GRH gives pointwise. Sparse needs pointwise; pointwise caps at $1/2$; dense forfeits the gain: the same square-root wall as Propositions 3.1 and 3.2, probed along yet another axis of approach and holding. Third, the machinery is dual-use. Built to construct Carmichael numbers, it acquires primes directly: Sections 2–3 close the route through proximity, and the pigeonhole at its heart is a prime-existence theorem outright.

5. EMPIRICAL RESULTS

The theorems above are consistent with what one actually observes. Normalising the distance from m to the nearest prime by $\log m$ (so a “random odd number” baseline is $\frac{1}{2}$ in the mean):

population	n	mean $d/\log m$	median
all Carmichael $m \leq 10^8$ (exact census)	255	0.550	0.454
matched random odd composites	12750	0.482	0.354
constructed Carmichael, 23–77 digits	1000	0.483	0.343
matched random odd	1000	0.494	0.359
matched random $\equiv 1 \pmod{720720}$	1000	0.508	0.360

Carmichael numbers are not closer to primes than random numbers of their size; at small scales they are, if anything, slightly farther. What their structure does control is *where* the nearby primes sit: writing $m \equiv 1 \pmod{L}$ with λ -structure as in [3], a candidate $m + c$ avoids every small prime $r \mid \lambda(L)$ unless $r \mid c + 1$. In all 1000 constructed cases the nearest prime occupied such an admissible offset ($\gcd(c \pm 1, 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13) = 1$), and a search that skips inadmissible offsets found a certified prime above 10^{40} in a median of 12 (maximum 152) Baillie–PSW tests per neighborhood over 1000 engagements, without a miss. This is a constant-factor convenience, exactly as the theory predicts: the Mertens boost $\prod_{r \mid \lambda} r/(r-1)$ on admissible offsets is cancelled, on average, by their sparsity. Scripts reproducing all computations in this section are available from the author.

6. RELATED WORK AND CLOSING REMARKS

Pintz, Steiger and Szemerédi [9] exhibited an infinite set of primes, of counting density $\gg x^{2/3}/\log x$, whose membership can be decided in deterministic polynomial time; but their *generation* procedure is randomized, and no deterministic procedure is known that produces an element of their set (or any prime) beyond a given bound n in time $n^{o(1)}$. The companion paper [3] achieves precisely this on the composite side, for Carmichael numbers; the present note quantifies why that success does not transfer to primes through proximity: the transfer is throttled by Carmichael *counting* (Theorem 2.2 and Corollary 2.3), capped below polynomial proximity by the Pomerance upper bound (Proposition 3.1), and in its algorithmic form defeated by the cost of counting certificates (Proposition 3.2).

We close by lasing the one target this analysis designates: *any improvement of Harman’s* $C(X) \geq X^{0.33}$ *past* $X^{0.706}$ *makes Corollary 2.3 nonvacuous*, yielding primes provably closer to (infinitely many) Carmichael numbers than they are known to be to anything else. Conversely, improvements in the gap moment (1) lower the 0.706 threshold toward the abundance conjecture’s $\theta \rightarrow 1$; the two literatures meet in the middle of this trade-off.

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